2.2 State of the Art of Survival Analysis in Healthcare

SA represents a key tool in the intersection of AI and healthcare, offering a statistical framework specifically designed to analyze and model time-to-event data. These 'events' could range from death and disease recurrence to recovery or other healthcare milestones. With the advent of EHRs, wearable technologies, and patient monitoring systems, the availability of rich, complex datasets has increased, enabling the application of SA to address pressing medical challenges. SA facilitates predictions of critical events such as disease progression and treatment outcomes but also aids in bridging statistical methodologies with real-world healthcare decision-making.

SA first appeared in the 17th century with the pioneering tables by John Graunt about mortality and life expectancy [40], which established the foundation for actuarial science and demographic studies. Initially focused on analyzing death rates and life spans, SA evolved over centuries to encompass various applications within biomedical research and fields like sociology, criminology, marketing, and institutional research. By the mid-20th century, advancements in statistical methodologies and computational capabilities allowed SA to expand its scope significantly. In healthcare, it became instrumental in evaluating the impact of interventions, understanding disease progression, and informing patient care strategies. Its importance has only grown with the integration of modern computational techniques, including AI, which enables the analysis of high-dimensional, heterogeneous, and often incomplete datasets.

The applications of SA in healthcare are extensive and diverse, playing a key role in prognosis, risk assessment, treatment evaluation, and operational planning. Prognostic modeling is one of its most prominent uses, where SA estimates survival probabilities and predicts the likelihood of events such as disease recurrence or death. For instance, survival curves, a core output of SA, enable oncologists to visualize and assess expected outcomes for cancer patients undergoing specific treatments. Similarly, SA supports risk stratification by identifying patients at varying levels of risk for adverse events. This stratification informs personalized care, such as guiding preventive strategies for individuals at high risk of cardiovascular disease.

In clinical research, SA is essential to evaluate treatments by comparing their effectiveness in controlled trials. For example, the Kaplan-Meier (KM) [41] survival curve and log-rank test are widely employed to determine differences in time-to-event outcomes between treatment groups. Furthermore, SA addresses complex scenarios involving CR, where multiple potential events (e.g., death from different causes) may occur. By accounting for interdependencies between these risks, SA provides accurate cause-specific risk assessments, which are critical for designing effective interventions and treatment plans.

Beyond individual patient care, SA informs predictions about the progression of chronic diseases. For instance, it models the time to significant milestones, such as kidney failure in chronic kidney disease or the onset of symptoms associated with Alzheimer's. This predictive capability is invaluable for long-term patient management. Additionally, SA contributes to healthcare operations by optimizing hospital resource allocation. SA supports efficient bed utilization and staffing strategies by modeling patient length-of-stay or predicting discharge times.

SA remains a cornerstone of modern healthcare analytics, empowering clinicians and researchers to extract meaningful insights from complex temporal data. Its integration with AI promises to expand its applicability further, making it a critical component of personalized medicine and healthcare optimization.

2.2.1 Fundamentals

The foundations of SA lie in its mathematical constructs, which enable robust modeling and interpretation of time-to-event data. These constructs include the survival function, hazard function, cumulative hazard, and considerations for censored data.

In a conventional time-to-event or SA setup, a dataset D consists of N observations. Each of these observations is described by triplets $D = (x_i, t_i, d_i)_{i=1}^{N}$, where $x_i = (x_i^1, ..., x_i^c, ..., x_i^{Cov})$ is an Cov-dimensional vector where $c = 1, 2, ..., Cov$ indexes the covariates, t_i is the time-to-event, and $d_i \in \{0, 1\}$ is the censoring indicator. When $d_i = 1$, the event of interest is observed at time t_i; conversely, $d_i = 0$ indicates that the event is censored and has not occurred within the observed time.

The primary objective of SA is to model the probability of event occurrence over time. Central to this are three mathematical constructs:

1. **Survival Function $S(t|x)$:** The survival function represents the probability that the event occurs after a given time t, conditional on the covariates x, and is defined as:

$$S(t|x) = P(T > t|x) = 1 - F(t|x), \tag{2.1}$$

   where $F(t|x)$ is the Cumulative Distribution Function (CDF) of the event times and T is the random variable representing time-to-event (we use capital T for the random variable and t for its realizations). The survival function monotonically decreases with t. This function provides a holistic view of survival probabilities over time, which is critical for prognosis and treatment planning.

2. **Hazard Function $h(t|x)$:** The hazard function quantifies the instantaneous rate of event occurrence at a specific time t, given survival up to that time:

$$h(t|x) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t | T \geq t, x)}{\Delta t}. \tag{2.2}$$

   The hazard function highlights periods of increased risk and is instrumental in identifying high-risk patient cohorts. Like $S(t|x)$, $h(t|x)$ is a non-negative function. While all the survival functions $S(t|x)$ decrease over time, the hazard function $h(t|x)$ can have a variety of shapes. *For ease of notation, we will drop the dependence of the survival and hazard functions with the covariates.*

   Using the relations between the survival and probability density functions, $h(t)$ can also be expressed as:

$$h(t) = \lim_{\Delta t \to 0} \frac{F(t + \Delta t) - F(t)}{\Delta t \cdot S(t)} = \frac{p(t)}{S(t)}, \tag{2.3}$$

where $p(t) = \frac{dF(t)}{dt}$ is the time Probability Density Function (pdf). The time pdf provides the density of the event times and encapsulates the interplay between the instantaneous risk and the probability of surviving up to that time as $p(t) = h(t)S(t)$.

3. **Cumulative Hazard Function $H(t)$:** $H(t)$ aggregates the hazard over time, providing a cumulative measure of risk:
$$H(t) = \int_0^t h(u)du. \tag{2.4}$$
The integral captures the total 'amount of risk' accumulated from the start of the observation to time t. While less commonly used directly in analysis, $H(t)$ establishes an essential link between the hazard and survival functions.

4. **Relations between $S(t)$, $H(t)$ and $h(t)$:** The cumulative hazard function $H(t)$ and the survival function $S(t)$ are connected as follows:
$$S(t) = e^{-H(t)}. \tag{2.5}$$
This relation arises from the definition of the hazard function and the survival function:
$$h(t) = -\frac{d}{dt} \ln S(t). \tag{2.6}$$
Integrating both sides from 0 to t gives:
$$\ln S(t) = -\int_0^t h(u)du = -H(t). \tag{2.7}$$
Exponentiating both sides yields:
$$S(t) = e^{-H(t)}. \tag{2.8}$$
This exponential relation illustrates how the cumulative hazard encapsulates the survival probability in a multiplicative decay framework.

Figure 2.4 visually illustrates the relationships between these functions, providing a conceptual understanding of their interplay.

Among these functions, $S(t)$ and $h(t)$ are the most widely used in practice due to their interpretability and direct applicability to clinical and operational decision-making. The cumulative hazard function $H(t)$, while mathematically significant, is less commonly used directly, as it primarily serves to bridge $h(t)$ and $S(t)$. The probability density function $p(t)$ is often a byproduct of these analyses, used primarily in model fitting and estimation.

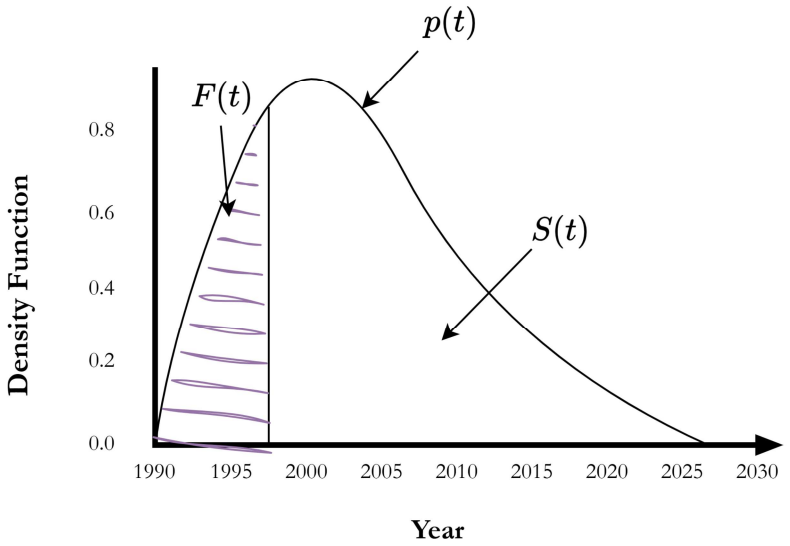


Figure 2.4: Relationships between key SA functions. Interplay between the survival function $S(t)$, the CDF $F(t)$, and the probability density function $p(t)$ in the context of time-to-event modeling.

A distinct feature of SA is its ability to handle censored data where the event of interest is not observed during the study period [42]. Censoring is ubiquitous in real-world healthcare scenarios, such as clinical trials, longitudinal studies, or chronic disease management, and arises due to various reasons [43]:

- **Right Censoring:** Occurs when the event time of a subject is unknown because they are lost to follow-up, withdraw from the study, or the study ends before the event occurs. For example, a patient surviving beyond the study period has a right-censored observation. This type of censoring is the most common form [44]. Mathematically, if a sample i is right-censored at time t_i, the true event time T_i is unknown but satisfies $T_i > t_i$.
- **Left Censoring:** Occurs when the event of interest has already occurred before the subject enters the study. For instance, patients diagnosed with a disease before enrollment may have a left-censored time-to-diagnosis. Mathematically, if a sample i is left-censored at time t_i, the true event time T_i is unknown but satisfies $T_i < t_i$.
- **Interval Censoring:** In this case, the exact event time is unknown but lies within a known interval. This is common in periodic follow-ups or scheduled clinical visits. Mathematically, for interval censoring, the true event time T_i satisfies $t_{i1} < T_i \geq t_{i2}$, where t_{i1} and t_{i2} are the bounds of the observed interval.

Figure 2.5 illustrates the different types of censoring commonly encountered in SA, highlighting the defined scenarios.

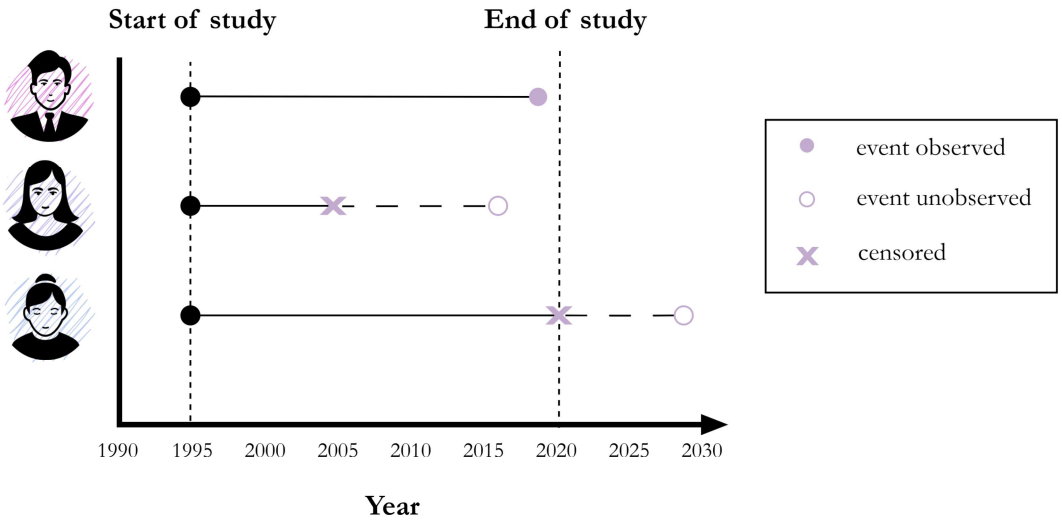


Figure 2.5: Timeline of a study with three patients showing SA outcomes. Solid circles indicate observed events, open circles indicate unobserved events, and 'X' marks denote censored data points. The timeline spans from the start to the end of the study.

2.2.2 Classical Approaches

SA has a rich history rooted in classical statistical methodologies that remain foundational for time-to-event data analysis. Among these, the KM estimator and Cox proportional hazards model (CoxPH) [45] provide robust tools for modeling and interpreting survival data. Despite the emergence of more complex ML and DL techniques, these classical approaches retain relevance due to their interpretability, simplicity, and effectiveness in many applications.

While both classical statistical methods and ML-based approaches share the target of predicting survival times and estimating survival probabilities, their emphases differ. Classical methods focus on characterizing the distributions of event times and the statistical properties of parameter estimation, typically visualized through survival curves. Conversely, ML approaches prioritize predictive accuracy, often leveraging the power of traditional SA frameworks in conjunction with advanced algorithms to improve the prediction of event occurrences at specific time points. This distinction underscores the complementary nature of traditional and modern methods in advancing the field of SA.

Classical statistical methods in SA can be classified into three categories based on the underlying assumptions and how model parameters are used: non-parametric, semi-parametric, and parametric.

Non-Parametric Methods

Non-parametric methods are particularly valuable in SA when no assumed underlying distribution for event times exists or when the proportional hazards assumption does not hold. These methods allow flexibility and adaptability in modeling survival data without imposing

restrictive assumptions.

Among these methods, the KM estimator remains the most widely used non-parametric approach to estimate the survival function $S(t)$, which represents the probability that an event of interest occurs after time t as defined previously. Introduced in 1958, the KM estimator [41] is particularly useful for datasets with censored observations, as it accounts for incomplete information about time-to-event data.

Let $T_1, T_2, T_3, \ldots, T_K$ denote the distinct ordered event times for the population of N observations. At a specific event time T_j, there are $\delta_j \geq 1$ observed events, and $risk_j$ individuals considered 'at risk', where $risk_j$ accounts for those who have not experienced the event or been censored before T_j. The conditional survival probability beyond T_j is given by:

$$p(T_j) = \frac{risk_j - \delta_j}{risk_j}. \tag{2.9}$$

The KM estimator of $S(t)$ is then defined as the product of these conditional probabilities:

$$\hat{S}(t) = \prod_{T_j \leq t} \left(1 - \frac{\delta_j}{risk_j}\right), \tag{2.10}$$

where the product accumulates overall event times T_j up to t.

The KM curve provides a stepwise estimate of the survival probability over time, with each step corresponding to an event. Its key advantages include:

1. **Non-parametric Nature:** The KM estimator does not require assumptions about the underlying distribution of survival times, making it highly flexible.
2. **Handling of Censoring:** The estimator accommodates right-censored data, ensuring accurate survival probability estimates even when some individuals are lost to follow-up.

However, this method presents two main limitations: (1) it cannot directly incorporate covariates to adjust for individual differences in risk factors, and (2) it assumes that censored individuals have the same survival probability as those who remain uncensored, which may not always hold.

The Nelson-Aalen (NA) estimator [46], [47], another non-parametric method, provides a cumulative hazard function estimate based on modern counting process techniques. It is particularly useful for analyzing the hazard of events over time. The NA estimator is defined as:

$$\hat{H}(t) = \sum_{T_j \leq t} \frac{\delta_j}{risk_j}, \tag{2.11}$$

where δ_j is the number of events that occurred at time T_j, and $risk_j$ is again the number of individuals at risk (not censored) before T_j. While less commonly used for direct survival probability estimation than the KM estimator, the NA method provides valuable insights into the event intensity over time. It serves as a basis for more advanced SA techniques.

The Life-Table (LT) method [48] is a variant of the KM estimator designed for interval-grouped survival data. Instead of estimating survival probabilities continuously, the LT

method aggregates data into time intervals, estimating survival probabilities for each interval based on grouped events and risk counts. This approach is particularly advantageous for large populations or datasets with interval-censored data. However, the interval-based nature of LT reduces precision compared to the continuous-time KM method.

These non-parametric methods form the foundation of SA, offering robust tools to estimate survival and hazard functions without relying on stringent distributional assumptions. Their adaptability makes them indispensable when flexibility and minimal assumptions are paramount.

Semi-Parametric Models: Cox Proportional Hazards Model

CoxPH [45], developed in 1972, is one of the most influential tools in SA. It provides a robust framework for evaluating the effects of multiple covariates on survival outcomes. As a hybrid between parametric and non-parametric approaches, CoxPH offers the flexibility of not requiring a predefined distribution for survival times while enabling precise estimation of covariate effects through the proportional hazards assumption. This balance makes CoxPH particularly valuable in clinical investigations where multiple interacting factors influence survival.

Unlike simpler methods like the KM estimator, which considers a single covariate at a time, the CoxPH model accommodates multivariate data, allowing researchers to analyze the effect of several covariates simultaneously. For instance, factors like age may also play a significant role when comparing survival outcomes for patients with different genotypes. CoxPH adjusts for these additional variables, providing a comprehensive estimate of the effect size of each covariate while controlling for confounding factors.

The CoxPH model specifies the hazard function $h(t)$, representing the instantaneous rate of event occurrence at time t, as:

$$h(t) = h_0(t) \exp(\beta^T x), \tag{2.12}$$

where $h_0(t)$ is called baseline hazard and represents the hazard value if all covariates are equal to zero (the quantity $\exp(0)$ equals 1), and β is the vector of regression coefficient quantifying the effect of covariates x on the hazard. The proportional hazards assumption ensures that the hazard ratio HR between any two individuals remains constant over time, independent of the baseline hazard. For two individuals with covariate vectors x_1 and x_2, the hazard ratio is given by:

$$HR = \frac{h(t|x_1)}{h(t|x_2)} = \exp(\beta^T(x_1 - x_2)). \tag{2.13}$$

This formulation emphasizes the relative risk of event occurrence between individuals while abstracting the absolute hazard rate.

The CoxPH model employs a semi-parametric approach, leaving $h_0(t)$ unspecified, which makes it more flexible than fully parametric models. The regression coefficients β are estimated using the partial likelihood method, which depends only on the observed event times and avoids

explicitly modeling the baseline hazard. For each covariate, the hazard ratio $HR = \exp(\beta)$ provides a direct interpretation:

- A hazard ratio of 1 ($\beta = 0$) indicates no effect of the covariate on the hazard.
- A hazard ratio greater than 1 ($\beta > 0$) signifies an increased hazard (i.e., a higher likelihood of the event and shorter survival time).
- A hazard ratio less than 1 ($\beta < 0$) denotes a decreased hazard (i.e., a lower likelihood of the event and longer survival time).

The survival function $S(t)$, which represents the probability of survival beyond time t, can be derived from the hazard function as:

$$S(t) = S_0(t)^{\exp(\beta^T x)}, \tag{2.14}$$

where $S_0(t) = \exp(-H_0(t))$ is the baseline survival function, and $H_0(t)$ is the cumulative baseline hazard function. The estimator introduced by Breslow [49] is commonly used to approximate $H_0(t)$, given by:

$$\hat{H}_0(t) = \sum_{t_i \leq t} \frac{1}{\sum_{j \in Risk_i} \exp(\beta^T x_j)}, \tag{2.15}$$

where $Risk_i$ denotes the set of individuals at risk at time t_i.

Extensions of the CoxPH model
Given the limitations of the basic CoxPH model, such as its reliance on the proportional hazards assumption, various extensions and adaptations have been developed:

- **Regularized Cox models:** These include Lasso-Cox [50], Ridge-Cox [51], and Elastic Net (EN) [52] Cox models, which incorporate penalty functions to handle high-dimensional data and select significant features while mitigating overfitting. For example, Lasso applies an l_1-norm penalty for sparse feature selection, while Ridge uses an l_2-norm for handling correlated features.
- **CoxBoost [53]:** This method applies gradient boosting to Cox regression, enabling the inclusion of mandatory covariates while improving prediction performance in high-dimensional settings.
- **Time-Dependent Cox Models [54]:** These models handle covariates that change over time, such as blood pressure or temperature, by modifying the hazard function to incorporate time-varying effects.

The CoxPH model is extensively used in healthcare to evaluate survival outcomes in diseases such as cancer, cardiovascular conditions, and chronic illnesses. It is particularly effective in assessing the impact of demographic, clinical, and genetic factors. However, the model has limitations, including its reliance on the proportional hazards assumption, which may not hold in all datasets. This has motivated the development of robust variants and extensions to ensure its continued applicability in modern SA.

Parametric Models

Parametric survival models provide a robust alternative to non-parametric (e.g., KM) and semi-parametric (e.g., CoxPH) methods by assuming that survival times, or their logarithmic transformations, follow a specific theoretical distribution, such as exponential, Weibull, log-normal, or log-logistic [43]. These models are particularly valuable when the assumed distribution aligns well with the underlying data, offering a simple and efficient framework for predicting survival probabilities and time-to-event outcomes.

Parametric models assume that the survival time T, or its logarithm $\ln(T)$, adheres to a known probability distribution. For instance:

- The Exponential distribution assumes a constant hazard rate over time, characterized by a single parameter λ. Its survival function is given by:

$$S(t) = \exp(-\lambda t), \tag{2.16}$$

  where $\lambda > 0$ is the rate parameter. This simplicity makes it ideal for scenarios where the event risk is time-invariant.

- The Weibull distribution, one of the most versatile parametric models, introduces a shape parameter $\alpha > 0$ to allow for varying hazard rates:

$$S(t) = \exp\Big(-\Big(\frac{t}{\lambda}\Big)^{\alpha}\Big), \tag{2.17}$$

  where $\lambda > 0$ is the scale parameter and $\alpha > 0$ determines the shape of the hazard function. If $\alpha = 1$, the Weibull model reduces to the exponential model. When $\alpha > 1$, the hazard rate increases over time, while $\alpha < 1$ implies a decreasing hazard rate.

- The Log-normal and Log-logistic distributions model survival times using transformations to log space, allowing for non-monotonic hazard functions. These distributions are particularly useful when hazard rates initially increase and decrease over time, capturing more complex survival patterns.

Parametric models utilize the Maximum Likelihood Estimation (MLE) technique to estimate model parameters. Suppose the dataset consists of N with v censored and $N - v$ uncensored observations. Let $\beta = (\beta_1, ..., \beta_P)^T$ denote the parameter vector. For a given instance i, if the event occurs at time T_i, the likelihood is driven by the probability density function $f(T_i, \beta)$. Conversely, if the instance is censored at time V_i, the likelihood incorporates the survival function $S(V_i, \beta)$, reflecting the probability that the event has not yet occurred.

The combined likelihood function for all instances is given by:

$$L(\beta) = \prod_{d_i=1} f(T_i, \beta) \prod_{d_i=0} S(V_i, \beta). \tag{2.18}$$

The MLE framework optimizes this likelihood to estimate the parameters β, enabling predictions for survival probabilities and hazard rates.

A special class of parametric models is the Accelerated Failure Time (AFT) model [54], which assumes a linear relationship between the logarithm of survival time and covariates:

$$\ln(T) = X\beta + \sigma\epsilon, \tag{2.19}$$

where $\sigma > 0$ is an unknown scale parameter, and ϵ is an error term following a specified distribution. AFT models are particularly appealing as they provide direct estimates of time acceleration or deceleration due to covariates, allowing for intuitive interpretations of how covariates influence the time to the event of interest.

Parametric survival models are widely used in healthcare, engineering, and reliability analysis due to their efficiency and ability to directly estimate survival probabilities, hazard rates, and median survival times. For example, the Weibull model is often applied in cancer prognosis studies, while the log-normal model analyzes treatment durations for chronic diseases.

However, the primary limitation of parametric models is their reliance on the assumed distribution. The resulting estimates can be biased and misleading if the chosen distribution does not align well with the underlying survival times. Non-parametric or semi-parametric methods are more appropriate for datasets with no theoretical distribution.

Despite these limitations, parametric models remain essential in SA. They offer a powerful and interpretable framework for understanding time-to-event data. Their mathematical rigor and computational efficiency combination ensures their continued relevance across diverse domains.

2.2.3 Machine Learning Approaches

ML has significantly advanced SA by addressing the limitations of traditional statistical models such as KM estimators and CoxPH models. While classical methods provide interpretable and foundational tools, they often struggle with high-dimensional, nonlinear, and multimodal data. ML methods, ranging from tree-based algorithms to DL architectures, enable the modeling of complex patterns in survival data and the integration of diverse data sources, such as genomic profiles, medical imaging, and EHRs.

Tree-Based Methods

Tree-based approaches are among the earliest ML methods adapted for SA. They extend decision tree algorithms to accommodate censored data, recursively partitioning the data based on covariates to model time-to-event outcomes.

- **Random Survival Forests (RSFs)** [55]: RSF improves upon survival trees by creating ensembles of survival trees using bootstrap aggregation (bagging) and random feature selection. This method enhances predictive accuracy and robustness while effectively handling complex interactions among covariates and right-censored data.
- **Gradient-Boosted Survival Trees (GBST)** [53]: GBST refines survival predictions by iteratively optimizing a survival-specific loss function, offering improved accuracy over single-tree models.

Bayesian Methods

Bayesian approaches provide a probabilistic framework for SA, integrating prior knowledge with observed data. Bayesian Networks (BNs) model dependencies among covariates, while

Naïve Bayes classifiers offer a simplified yet effective probabilistic approach to survival prediction [56]. These methods excel in scenarios with limited data or where uncertainty quantification is critical, such as clinical decision-making.

Support Vector Machines

Support Vector Machines (SVMs) have been adapted for SA by incorporating methods to handle censored data. For example, Support Vector Regression for Censored Data (SVRc) [57] modifies the traditional SVM loss function to account for censored observations, enabling robust predictions in high-dimensional spaces. However, these methods may face computational challenges with large datasets due to their quadratic time complexity.

Deep Learning

DL methods have transformed SA by introducing models that capture complex, nonlinear relationships in high-dimensional datasets. They provide increased flexibility and predictive accuracy compared to traditional approaches. The key DL contributions to SA include:

- **Cox-Based Extensions:** The integration of Neural Networks (NNs) with CoxPH models began with the Faraggi and Simon model [58], which introduced a feed-forward NN to replace the linear predictor of the CoxPH model, enabling the capture of nonlinear relationships between covariates and hazard risks. Building on this foundation:
  - **DeepSurv [59]**: Extends the Faraggi model by parameterizing the log-risk function with a deep feed-forward NN, enabling the modeling of nonlinear covariate relationships while maintaining the proportional hazards assumption. Optimized using stochastic gradient descent, DeepSurv is particularly effective in large datasets with complex, nonlinear interactions.
  - **Cox-Time [60]**: Introduces time-varying effects by allowing the hazard function to depend on both covariates and time. Relaxing the proportional hazards assumption offers greater flexibility but increases computational demands.
  - **Cox-nnet [61]** and **Cox-PASNet [62]**: These models adapt DeepSurv for high-dimensional data, such as genomics, using regularization techniques to prevent overfitting.
  - **Luck's Model [63]:** Proposes an extension of DeepSurv with additional regularization techniques to improve interpretability and robustness in high-dimensional datasets, particularly in genomics and imaging data.
- **Discrete-Time Models:**
  - **DeepHit [64]:** A groundbreaking model that directly estimates the survival distribution without relying on proportional hazards or specific parametric assumptions. Its architecture includes a shared sub-network for learning general representations and additional task-specific sub-networks for survival predictions. DeepHit is particularly suited for modeling nonlinear relationships between covariates and survival times.

  - **Dynamic-DeepHit [65]:** Builds on DeepHit by incorporating RNNs to handle longitudinal data with time-varying covariates. This model captures temporal dependencies and includes an auxiliary prediction loss to enhance its ability to predict future covariate values, making it ideal for time-series survival datasets.

- **Fully Parametric Models:** DL has also advanced fully parametric survival models by employing architectures such as RNNs to predict survival distributions. RNN-Surv [66] sequentially predicts survival distributions using RNNs, capturing dependencies across time. It is particularly effective in longitudinal datasets with time-varying covariates. Weibull Time-to-Event RNN [67], based on the Weibull distribution: this model offers flexibility in estimating survival times while preserving parametric assumptions about the hazard function.

ML, particularly DL, has revolutionized SA by enabling flexible, accurate predictions in high-dimensional and multimodal data settings. While traditional tree-based and Bayesian methods remain valuable for interpretability and computational simplicity, DL approaches like DeepSurv, DeepHit, and Dynamic-DeepHit provide unparalleled capacity to model nonlinear and time-varying relationships, making them indispensable tools in precision medicine and beyond.

2.2.4 Competing Risks

Building upon SA advancements, CR represents a natural extension of traditional single-risk models to address more complex and realistic scenarios. While state-of-the-art models like DeepSurv [59] and Cox-Time [60] have achieved remarkable success in estimating survival probabilities and hazard functions, these approaches are limited to single-risk frameworks. In contrast, CR scenarios involve multiple mutually exclusive events, where the occurrence of one event precludes the observation of others. Addressing CR scenarios requires specialized models that capture the interplay between risks and estimate the probability of each specific event. This challenge remains underexplored in the current landscape of SA research.

In traditional SA, the focus is on estimating the survival function $S(t)$, which represents the probability of an individual surviving beyond time t. The survival function operates under the assumption of a singular event of interest, with censoring accounting for the absence of an observed event or loss to follow-up. However, this framework becomes insufficient in CR settings, where individuals are exposed to competing events. For example, in a medical study, a patient may die due to cardiovascular disease or cancer, but the occurrence of one event precludes the observation of the other.

To accommodate these complexities, the data representation changes. Each instance is now characterized as (x_i, t_i, r_i), where r_i denotes the event type:

- $r_i = 0$: The instance is censored.
- $r_i = k$ ($k \in \{1, 2, ..., R\}$): Event k occurred, where R is the total number of possible events.

This expanded notation emphasizes the need to model the probability of each specific event

while accounting for the competing nature of other risks.

The Cumulative Incidence Function (CIF) replaces the survival function as the primary modeling target in CR scenarios. The CIF for event k is defined as:

$$CIF_k(t|x) = P(T \leq t, r = k|x), \tag{2.20}$$

where T is the time-to-event random variable, and $r = k$ specifies that event k occurred. Unlike the survival function, which aggregates all risks into a single metric, the CIF explicitly disentangles the probability of each specific event. Key properties of the CIF include:

1. The CIF for any event k does not reach 1 as $t \to \infty$ due to the presence of CR.
2. The sum of CIFs over all events gives the probability of experiencing any event by time t:
$$\sum_{k=1}^{R} CIF_k(t|x) = P(T \leq t|x), \tag{2.21}$$
which excludes censoring.

Classical Methods for CR: The Fine-Gray Model

The Fine-Gray (FG) model [68] is one of the most widely used classical approaches for CR. FG directly models the subdistribution hazard for each event type, allowing for estimating the CIF without explicitly modeling the cause-specific hazards. It achieves this by reformulating the hazard function to reflect the probability of an event occurring in the presence of CR. While effective, the FG model, like the Cox proportional hazards model, relies on strong assumptions about the underlying stochastic processes and the form of the hazard function. These assumptions, such as proportional hazards, often fail to align with the complexities of real-world data, motivating the need for more flexible, data-driven methods.

Deep Learning Approaches to CR

The emergence of DL in SA has led to significant advancements in addressing CR. While most DL-based models focus on single-risk scenarios, a few state-of-the-art methods explicitly incorporate CR into their frameworks. These include:

- **DeepHit** [64]: DeepHit is a foundational model in DL-based CR SA, directly estimating the CIF for each competing event. Already defined for single-event SA, DeepHit can also be extended to handle CR. Its architecture features a shared sub-network to learn general latent representations from covariates and cause-specific sub-networks that model CIFs for each competing event. DeepHit operates discretely and uses a composite loss function combining likelihood-based terms and a ranking loss to improve the C-index. While highly flexible, its discrete-time framework requires time discretization, which can lead to information loss. Moreover, DeepHit produces numerical CIF estimates without analytical expressions, limiting statistical computations and precision.
- **DeepComp** [69]: DeepComp extends the discrete-time approach by employing RNN-based cause-specific sub-networks. Unlike DeepHit, DeepComp outputs cause specific

discrete hazards for each time interval rather than CIFs. These hazards can be aggregated to estimate the cumulative incidence of each risk. RNNs enable DeepComp to capture temporal dependencies more effectively than feed-forward networks. However, the reliance on discrete hazards shares similar limitations with DeepHit regarding continuous-time modeling and precision.

- **CRESA** [70]: The CR and Recurrent-Event Survival Analysis (CRESA) model also employs RNN architectures and discrete-time modeling. Like DeepHit, it generates a final distribution over all CR using a loss function based on recurrent CIFs. Additionally, it incorporates a ranking component to improve performance on metrics like the C-index. The RNN-based design from CRESA is particularly advantageous for datasets with sequential or recurrent events, although its discrete-time framework imposes similar constraints to DeepHit and DeepComp.
- **SSMTL** [71]: Semi-Supervised Multi-Task Learning (SSMTL) transforms CR SA into a multi-task classification problem. Each time point (or competing event) is treated as a separate binary (or multi-class) classification task. SSMTL uses a custom composite loss function that includes a classification loss for uncensored data, a semi-supervised loss for censored data, and regularization losses. While innovative, SSMTL operates in discrete time, similar to DeepHit, and its reliance on binary classification tasks limits the granularity of its predictions.
- **DeepCompete** [72]: DeepCompete introduces a novel continuous-time approach for CR modeling. It employs neural ordinary differential equation (nODE) blocks within each cause-specific sub-networks cause-specific sub-network, which outputs cumulative hazard functions. Unlike discrete-time methods like DeepHit and CRESA, the continuous-time framework of DeepCompete allows for more precise modeling of survival times and CIFs. However, using nODE blocks increases the computational complexity of the model.

Despite these advancements, modeling CR remains challenging due to disentangling overlapping risks and effectively managing censoring. Current state-of-the-art methods often require large datasets and computational resources, which may limit their applicability in real-world scenarios. Future research should focus on developing scalable and interpretable models that integrate CR seamlessly into SA frameworks while addressing the limitations of discrete-time and proportional hazard assumptions.

CR highlights the complexities of real-world SA, necessitating specialized models that go beyond traditional single-risk frameworks. By leveraging the flexibility and capacity of DL, researchers are making significant strides in this domain, with methods like DeepHit paving the way for more comprehensive CR modeling. These innovations hold the potential to transform applications ranging from personalized medicine to risk management, where understanding the interplay of competing events is critical.

2.2.5 Validation Techniques

Model evaluation is a critical component of SA, as it provides insights into the performance and reliability of predictive models, especially in the presence of censored data and varying time horizons. Unlike traditional regression or classification tasks, SA involves the dual challenge of handling incomplete data due to censoring and accurately predicting time-to-event outcomes. Effective validation techniques assess model accuracy and ensure robustness across diverse datasets and scenarios. The evaluation process typically focuses on two key aspects: discrimination (the ability to rank individuals by risk) and calibration (the agreement between predicted probabilities and observed outcomes).

As outlined previously, each dataset is characterized by triplets $D = (x_i, t_i, d_i)_{i=1}^{N}$, where x_i represents the covariate vector, t_i denotes the time-to-event, and $d_i \in \{0, 1\}$ indicates the censoring status.

The Concordance Index (C-index) is one of the most widely used metrics to evaluate the discriminatory power of a survival model. It measures the rank correlation between predicted risks and observed event times, offering an intuitive understanding of whether higher predicted risks correspond to shorter observed survival times. The general formulation of the C-index is:

$$C_{index} = P\Big(\hat{F}(t|x_i) > \hat{F}(t|x_j)\Big| d_i = 1, t_i < t_j, t_i \leq t\Big), \tag{2.22}$$

where $\hat{F}(t|x_i)$ is the estimated CDF for the covariates x_i at time t. The C-index evaluates pairwise concordance among comparable samples, emphasizing the ability of the model to rank correctly.

The C-index is adapted in CR settings to evaluate the concordance of the predicted CIF for specific competing events. It considers the predicted CIF of the event of interest at a given time, compared against those of other individuals who have not yet experienced the event:

$$C_{index} \approx \frac{\sum_{i \neq j} A_{k,i,j} \cdot \mathbb{I}\Big(CIF_k(t|x_i) \leq CIF_k(t|x_j)\Big)}{\sum_{i \neq j} A_{k,i,j}}, \tag{2.23}$$

where $A_{k,i,j}$ is the indicator function for comparable pairs (i, j) for event k. This formulation evaluates the ability of the model to rank correctly for each specific competing event, considering the unique challenges introduced by multiple outcomes.

However, the original C-index [73] assumes time-invariant risk and cannot accommodate dynamic changes. To address this limitation, the time-dependent C-index [74] extends the metric by accounting for evolving risks.

The Brier Score (BS) evaluates the squared error between predicted survival probabilities and observed outcomes. It incorporates the Inverse Probability of Censoring Weighting (IPCW) [75] to adjust for censored observations, assigning higher weights to uncensored instances. The BS at a specific time t is given by:

$$BS(t) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(S(t|x_i))^2}{G(t_i)}\cdot\mathbb{I}(t_i < t, d_i = 1) + \frac{(1-S(t|x_i))^2}{G(t)}\cdot\mathbb{I}(t_i \geq t)\right], \tag{2.24}$$

where $G(t)$ is the survival function for censoring.

The BS measures the ability of the model to rank individuals accurately by risk (discrimination) and the alignment of predicted survival probabilities with observed event frequencies (calibration).

To provide a time-agnostic assessment, the Integrated Brier Score (IBS) calculates the average BS over a range of times:

$$IBS(t_{max}) = \frac{1}{t_{max}}\int_0^{t_{max}} BS(t)dt. \tag{2.25}$$

The IBS is particularly valuable for comparing models across varying time horizons and assessing their overall predictive performance.

In CR settings, the BS and IBS are calculated for the CIF of each competing event k, providing a measure of the discrepancy between predicted and observed CIFs over time:

$$BS_k(t) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1-CIF_k(t|x_i))^2}{G(t_i)}\cdot\mathbb{I}(t_i < t, d_i = k) + \frac{(CIF_k(t|x_i))^2}{G(t)}\cdot\mathbb{I}(t_i \geq t)\right]. \tag{2.26}$$

The IBS for CR iBS_k is then calculated as:

$$iBS_k(t_{max}) = \frac{1}{t_{max}}\int_0^{t_{max}} BS_k(t)dt. \tag{2.27}$$

This metric evaluates the overall alignment between predicted and observed CIFs, accounting for the inherent complexity of multiple events.

Time-dependent metrics, like the time-dependent C-index and time-dependent BS, are essential for evaluating models that account for changes in risk over time. These metrics are especially relevant for modern survival models, such as RNNs and dynamic survival models, which explicitly model time-varying covariates. Time-dependent metrics ensure the evaluation reflects the dynamic nature of risks and outcomes in longitudinal datasets.

Censoring introduces unique challenges in SA, as it prevents the direct observation of event times for all individuals. This complicates the calculation of traditional metrics and necessitates specialized techniques like IPCW to handle censored observations. Key challenges include:

- **Biased Estimates:** Without proper adjustments, censored data can lead to biased model evaluations.
- **Comparable Pairs:** Metrics like the C-index rely on comparing pairs of individuals. Censored data reduce the number of comparable pairs, potentially lowering metric reliability.

- **Extrapolation:** For heavily censored datasets, models may overfit or extrapolate survival probabilities inaccurately, leading to unreliable calibration.

SA validation techniques ensure that models perform well in realistic scenarios characterized by censored data and dynamic risk profiles. Metrics like the time-dependent C-index and IBS provide comprehensive insights into the discrimination and calibration capabilities of a survival model. However, the presence of censoring necessitates thoughtful adaptations and robust statistical techniques to ensure unbiased and reliable evaluation. As SA models become more sophisticated, leveraging advanced metrics and addressing challenges with censored data will remain pivotal in advancing the field.

2.2.6 Challenges and Future Directions

Despite the significant progress in SA, particularly with the integration of ML and DL, several challenges and research gaps remain. These challenges, summarized in Figure 2.6, include handling high-dimensional data, modeling complex relationships, addressing censoring, and ensuring flexibility in training. Additionally, advancing methods for CR and incorporating parametric models into DL frameworks are critical areas for future research.

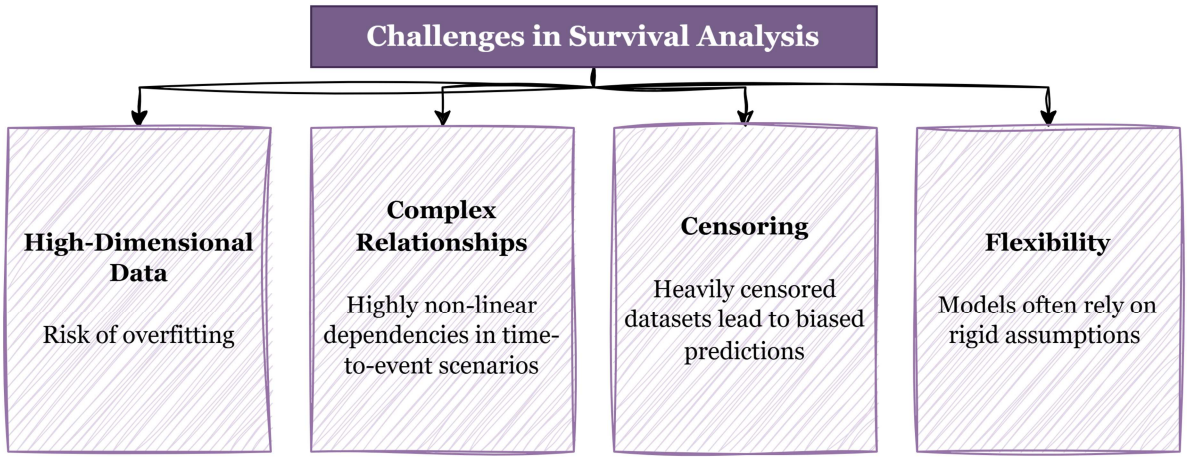


Figure 2.6: Challenges in Survival Analysis. A summary of key limitations, including handling high-dimensional data, modeling complex relationships, addressing censoring, and ensuring flexibility in training.

Current Limitations

1. **High-Dimensional Data:** Modern SA applications, such as those in genomics and medical imaging, often involve datasets with numerous covariates relative to the sample size. DL models, while powerful, can struggle with overfitting in such settings. This limitation necessitates robust regularization techniques and effective feature selection to ensure meaningful and reliable model performance.
2. **Complex Relationships Between Covariates:** DL models are adept at capturing intricate relationships within data; however, modeling highly non-linear and intricate

dependencies, especially in time-to-event scenarios, remains challenging. Ensuring these models adequately capture interactions without introducing biases or artifacts is difficult.

3. **Censoring:** Heavily censored datasets reduce the amount of observable information for model training, leading to biased predictions and reduced reliability of validation metrics. Current approaches, such as IPCW, offer partial solutions but are not universally robust.
4. **Flexibility:** Current approaches often require specific assumptions about time distributions or pre-defined functional forms, reducing their adaptability to diverse data settings. Flexible models that do not rely on rigid time assumptions are crucial for broad applicability across real-world scenarios.

Research Gaps

1. **Integration of CR:** Although CR modeling has seen notable advancements in DL-based SA, current methods often struggle to integrate multiple competing events into a unified framework seamlessly. Future research should focus on developing more effective architectures for CR modeling that balance accuracy, interpretability, and computational efficiency.
2. **Parametric Models with DL:** The integration of parametric survival models within DL architectures is underexplored. Parametric models offer interpretable survival estimates and flexible, functional forms, but combining these strengths with the representation learning capabilities of DL could unlock powerful hybrid approaches for SA. Future work should design methods that effectively merge these paradigms, ensuring flexibility and interpretability.

Addressing these challenges requires a multidisciplinary approach combining optimization innovations, neural architecture design, and statistical theory. Future efforts should prioritize methods that balance interpretability with adaptability, ensuring robustness in real-world, complex datasets. By addressing these limitations and research gaps, SA can become a reliable tool for critical healthcare domains.

2.2.7 State-of-the-Art Models Summary

The development of SA methods spans classical, ML, and DL paradigms, each offering unique advantages and limitations. Classical methods like the KM estimator and CoxPH provide interpretable and foundational tools for modeling time-to-event data. ML approaches like RSF and Bayesian models enhance predictive capabilities by capturing complex patterns in the data. DL methods, including DeepSurv and Dynamic-DeepHit, further expand the modeling capacity to handle nonlinear relationships, high-dimensional covariates, and CR.

Table 2.2 provides a comprehensive summary of key SA models discussed in Section 2.2. The table categorizes the models into classical, ML, and DL approaches while detailing their type (e.g., parametric, non-parametric, semi-parametric), support for CR, covariate handling, and distinguishing features. This overview highlights the progression from interpretable statistical models to advanced neural networks capable of addressing the complexities of real-world datasets.

Category	Model	Type	CR Support	Covariate Handling	Key Features
Classical	KM estimator [41]	Non-parametric	No	None	Stepwise survival estimation
	NA estimator [46], [47]	Non-parametric	No	None	Cumulative hazard estimation
	LT estimator [48]	Non-parametric	No	None	Interval-grouped survival estimation
	CoxPH [45]	Semi-parametric	No	Multivariate	Proportional hazards; interpretable hazard ratios
	Cox Regularized (Lasso [50], Ridge [51], EN [52])	Semi-parametric	No	High-dimensional	Handles correlated/large covariates with penalties
	CoxBoost [53]	Semi-parametric	No	Multivariate	Gradient boosting; feature selection
	TD-Cox [54]	Semi-parametric	No	Time-varying covariates	Models dynamic risk factors
	FG [68]	Semi-parametric	Yes	Multivariate	Subdistribution hazards; CR-specific
	Exponential [43]	Parametric	No	Multivariate	Constant hazard; simple and interpretable
	Weibull [43]	Parametric	No	Multivariate	Flexible hazard (increasing/ /decreasing) widely used
Machine Learning	RSF [55]	Non-parametric	No	Multivariate	Ensemble of trees; handles censoring
	GBST [53]	Non-parametric	No	Multivariate	Survival-specific loss function
	Bayesian Methods [56] (BN, Naïve Bayes)	Parametric	No	Multivariate	Probabilistic modeling; quantifies uncertainty
	SVRc [57]	Non-parametric	No	Multivariate	Adapts SVMs for censored data
Deep Learning	Faraggi and Simon's model [58]	Semi-parametric	No	Multivariate	First DL extension of CoxPH; captures nonlinear effects
	DeepSurv [59]	Semi-parametric	No	Multivariate	Nonlinear relationships; proportional hazards
	Luck's Model [63]	Semi-parametric	No	High-dimensional	Improved interpretability for genomics data
	Cox-Time [60]	Semi-parametric	No	Multivariate + Time	Time-varying hazards
	Cox-nnet [61] and Cox-PASNet [62]	Semi-parametric	No	High-dimensional	Optimized for high-dimensional genomic data
	DeepHit [64]	Non-parametric	Yes	Multivariate	Direct CIF estimation; discrete-time modeling
	Dynamic-DeepHit [65]	Non-parametric	Yes	Time-varying covariates	RNN-based temporal dependencies; CR modeling
	DeepComp [69]	Non-parametric	Yes	Multivariate	Cause-specific discrete hazards; RNN-based
	CRESA [70]	Non-parametric	Yes	Multivariate	Recurrent CIFs; RNN for sequential data
	SSMTL [71]	Non-parametric	Yes	Multivariate	Semi-supervised; multi-task classification
	DeepCompete [72]	Non-parametric	Yes	Multivariate	Continuous-time CR with nODE blocks

Table 2.2: Comparison of SA models described in Section 2.2. The table summarizes key models for SA, indicating their type, CR support, covariate handling, and distinguishing features.